

PROPAGATION AND MITIGATION OF PROPAGANDA IN SOCIAL NETWORKS

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DECLARATION BY STUDENTS

We **Ritika Singh, Shireen Kachroo and Paakhi Somani** declare that this submission is our own work and that, to the best of our knowledge and belief, it contains no material previously written by another person nor material which has been accepted for the award of any other degree or diploma of the university or other institute of higher learning, except where due acknowledgement has been made in the text.

1. List of Pre Built Libraries: NetworkX, Pyvis, pandas, matplotlib, scikit-learn, transformers/torch
2. List of Pre Built Features: TF-IDF text vectorization, NetworkX graph generation, Centrality Calculation, ML classifiers
3. Percentage of pre written source code and code written by us: 20%, 80%

Place: Noida, India

Date: May 4,2026

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DECLARATION BY SUPERVISOR

I, **Dr. Ankita Verma**, declare that the above-submitted project titled '**Propagation and Mitigation of Propaganda in Social Networks**' was conducted under my supervision. The project is original, not copied from any external source, and has not been submitted earlier in JIIT.

Remarks (if any):

Signature (Supervisor)

CERTIFICATE

This is to certify that the work titled '**Propagation and Mitigation of Propaganda in Social Networks**' submitted by **Ritika Singh, Shireen Kachroo** and **Paakhi Somani** in partial fulfilment for the award of degree of **B.Tech** in **Computer Science** of Jaypee Institute of Information Technology, Noida has been carried out in my supervision. This work has not been submitted partially or wholly to another University or Institute for the award of this or other degree or diploma.

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Designation: Assistant Professor (Senior Grade)

Date: May 4, 2026

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ABSTRACT

This project focuses on understanding how propaganda spreads on online social networks and how it can be controlled. It combines machine learning and social network analysis to study both the detection and the spread of propaganda. The system mainly performs two tasks. First, it identifies whether a given piece of information is propaganda or not using natural language processing and machine learning techniques. Second, it studies how propaganda spreads among users by using a simulation model.

For detecting propaganda, the system first cleans and prepares the text data. Techniques like TF-IDF and advanced language models such as BERT and RoBERTa are used to analyze the text in a better way. Different models are then trained, including Logistic Regression and deep learning models like BERT and RoBERTa. These models help in accurately classifying content, even when the data is noisy or changes over time.

To understand how propaganda spreads, the project uses concepts from social network analysis along with the SI (Susceptible–Infected) model. In this model, users are treated as nodes in a network. A user can first be unaware (susceptible), then start sharing propaganda (infected). The system also studies important parts of the network such as highly connected users (hubs), connecting nodes (bridges), and points where information flow slows down.

By combining detection and simulation, this project helps in visualizing how propaganda starts, how it spreads through the network, and which users play a major role in spreading it. It also shows how certain actions can help in reducing its spread. Even though online behavior can be unpredictable, this project shows that using machine learning and network analysis together can be useful in studying and controlling propaganda in social networks.

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LIST OF ABBREVIATIONS

NLP – Natural Language Processing

ML – Machine Learning

DL – Deep Learning

SNA – Social Network Analysis

SI Model – Susceptible–Infected Model

TF–IDF – Term Frequency–Inverse Document Frequency

XAI – Explainable Artificial Intelligence

BA Model – Barabási–Albert Model

LR – Logistic Regression

BERT – Bidirectional Encoder Representations from Transformers

RoBERTa – Robustly Optimized BERT Pretraining Approach

UAT – User Acceptance Testing

CPU – Central Processing Unit

GPU – Graphics Processing Unit

Chapter 1: Introduction

1. Background

In today's digital world, propaganda spreads very quickly through social media platforms. With millions of users connected in large networks, even a small piece of misleading or biased content can reach a wide audience in a short time. This makes it important to understand how propaganda starts, how it spreads, and how it influences users online.

This project aims to build a tool for studying the propagation of propaganda in online social networks. It is mainly designed for learning and analysis purposes, similar to simulation tools that are used to understand real-world systems. The project combines two main parts: a machine learning-based model for detecting propaganda, and a network simulation model to study how it spreads among users.

For detection, the system uses natural language processing (NLP) techniques along with models like Logistic Regression, BERT and RoBERTa to classify whether a given content is propaganda or not. At the same time, the spread of propaganda is modeled using the SI (Susceptible–Infected) model. In this model, users are considered as nodes in a network, where they move from being unaware (susceptible) to spreading propaganda (infected).

The simulation helps in visualizing how propaganda moves through different parts of a network, such as highly connected users (hubs) and connecting nodes (bridges). Overall, this project shows how machine learning and social network analysis can be combined to better understand and study the behavior of propaganda in online environments.

2. Problem Statement

Even though many machine learning and NLP techniques have been developed to detect propaganda, understanding how propaganda spreads in online social networks is still a major challenge. Social networks are highly dynamic, and user behavior keeps changing, which makes it difficult to study the flow of propaganda in a clear and structured way.

One of the main issues is the high spread rate of propaganda. Content that is emotionally appealing or controversial is shared more quickly by users, allowing propaganda to reach a large audience in a short time.

Another challenge is the lack of awareness among users. Many users are not able to identify whether the content they see is propaganda or not. This increases their chances of believing and further spreading such content.

The structure of social networks also makes the problem more complex. Some users act as hubs, meaning they have a large number of connections, while others act as bridges between different communities. These nodes play a major role in accelerating the spread of propaganda across the network.

In addition to this, most existing systems focus only on detecting propaganda in content. They do not provide tools to simulate how propaganda spreads or to identify which users contribute the most to its propagation.

This project aims to address these issues by combining machine learning-based propaganda detection with network-based spread modeling. The goal is to create a tool that not only identifies propaganda but also helps in understanding how it spreads within a network.

3. Social and Environmental Relevance

The spread of propaganda has significant social consequences, including manipulation of public opinion, polarization of communities, and disruption of democratic processes. A system capable of detecting and simulating propaganda spread can contribute to:

- Early identification of harmful information cascades
- Improved moderation strategies for social media platforms
- Increased awareness of how propaganda propagates in networks

From a broader societal perspective, such systems can assist policymakers, researchers, and platform administrators in designing safer and more transparent digital ecosystems.

Although the project is computational in nature, it also indirectly contributes to environmental sustainability by enabling more efficient detection mechanisms, reducing redundant manual moderation efforts, and optimizing computational resources through ensemble-based intelligent classification.

4. Gantt Chart/ Timelines

The development of the project was carried out in a structured phased manner to ensure systematic progress across all components:

Table 1: Project Timeline

Activity	Duration
Literature review on propaganda detection, SNA, and SI model	Jan 19 – Jan 25
Dataset collection, translation, preprocessing, and analysis	Jan 27 – Jan 29
Model development (Logistic Regression, BERT, RoBERTa)	Feb 25 – Mar 7
Ensemble model construction	Mar 10 – Mar 12

Network modeling + SI simulation	Apr 1 – Apr 3
Visualization using PyVis	Apr 5 – Apr 9
Mitigation strategy design & testing	Apr 12 – Apr 17
Integration, evaluation & report writing	Apr 28 – Apr 30

5. Scope

The scope of this project is to develop an integrated, simulation-based analytical tool for studying the propagation and mitigation of propaganda in online social networks. The system is designed to combine techniques from machine learning, deep learning, and network science in order to provide both content-level classification and structural propagation analysis within a unified framework.

At the core of the system lies a propaganda detection module, which classifies textual content using multiple models, including Logistic Regression, BERT, and RoBERTa. These models are combined through a soft voting ensemble approach, which improves overall performance by leveraging the strengths of both traditional and transformer-based methods. This ensures that the system can effectively handle different types of textual inputs, ranging from simple statements to more complex, context-dependent content.

Beyond classification, the project extends its scope to include network-based propagation modeling. Using graph-based representations, the system simulates how propaganda spreads across a social network, where users are modeled as nodes and their interactions as edges. The spread of propaganda is analyzed using the SI (Susceptible–Infected) model, enabling users to

observe how information diffusion evolves over time under different conditions.

A key feature of the system is its interactive visualization capability, implemented using NetworkX and PyVis. This allows users to dynamically observe the propagation process, where nodes change states based on their level of exposure. Such visualization enhances understanding by providing both a visual and analytical perspective of the spread.

The system also incorporates mitigation strategies to study how the spread of propaganda can be controlled. These include reducing the infection probability (β) of influential nodes, as well as deactivating hub nodes and bridge nodes to limit both rapid and cross-community propagation. Through these strategies, the system enables users to analyze the effectiveness of different intervention techniques.

Overall, the project functions as a simulation and educational tool, similar in concept to platforms like Gephi, allowing users to explore and understand the dynamics of propaganda spread in a controlled environment. While the system is primarily designed for academic and research purposes, it also provides a foundation for future extensions toward real-world applications in propaganda analysis and control.

Chapter 2: Literature Survey

1. Overview of the Existing Work/ Literature Review

In the current digital ecosystem, the rapid growth of online social networks has significantly transformed the way information is generated, shared, and consumed. While this connectivity has enabled efficient communication, it has also led to the widespread propagation of propaganda. As a result, detecting and controlling such content has become a major research focus in the fields of Natural Language Processing (NLP) and Social Network Analysis (SNA).

Existing research can broadly be categorized into three major directions:

1. Text-based propaganda detection using machine learning and deep learning models
2. Graph-based social network analysis for understanding influence and propaganda flow
3. Epidemic-inspired diffusion models for simulating propagation behavior

Traditional machine learning approaches such as Logistic Regression, Support Vector Machines, and Random Forest have been widely used for classification tasks. These models typically rely on handcrafted features such as TF-IDF representations, and lexical patterns. While effective in simpler scenarios, they often fail to capture deep contextual and semantic relationships in complex propaganda content.

To overcome these limitations, transformer-based models such as BERT and RoBERTa have been introduced in recent research. These models leverage attention mechanisms to understand contextual dependencies in text, making them highly effective for detecting subtle propaganda techniques.

In parallel, Social Network Analysis has been used to understand how information spreads across interconnected users. Graph-based metrics such as degree centrality, betweenness centrality, and community structure help in identifying influential nodes like hubs and bridges, which play a critical role in accelerating or controlling information diffusion.

Additionally, epidemic-based models such as the SI (Susceptible–Infected) model are widely used to simulate information spread. In this model, users transition from a susceptible state to an

infected state, representing adoption or sharing of information. This approach helps in understanding propagation dynamics and the structural impact of networks on diffusion.

2. Integrated Summary of the Literature Studied

Table 2: Summary Table of the Literature Studied

S.No	Article	Description	Metrics Used	Gaps / Shortcomings	Remarks
[1]	Detection of Propaganda in Information Warfare using Deep Learning (Pandey et al., IEEE, 2022)	Proposes a deep learning based approach to detect propaganda content on social media platforms. The model analyzes textual patterns and semantic structure to identify whether a piece of content is propagandistic in nature.	Accuracy, Precision, Recall, F1-Score	The work mainly focuses on text-level detection and does not study how propaganda spreads across network structures.	Provides a strong foundation for the propaganda classification component used in this project.

[2]	IEEE Access: New Techniques for Limiting Misinformation Propagation (2023)	This study focuses on reducing the spread of harmful content in social networks by applying intervention strategies such as limiting node influence and modifying network connections. .	Graph-based techniques, edge removal strategies, and influence control mechanisms.	The work focuses only on mitigation strategies and does not include content classification. .	Useful for designing intervention strategies in the SNA simulation.
[3]	SemEval-2020 Task 11: Detection of Propaganda Techniques in News Articles	This shared task focuses on identifying propaganda techniques in news articles using NLP methods. It provides annotated datasets and benchmarks for propaganda	F1-score, Precision, Recall	Primarily focused on text classification; does not consider network-based propagation.	Provides dataset and benchmarking support for training and evaluating the propaganda detection models.

		detection.			
[4]	Understanding Information Spread in Social Networks (Dash et al., WebSci)	Studies how information spreads across online social networks by analyzing user interactions and network structure. It highlights how certain nodes play a key role in accelerating or controlling the flow of information across communities.	Degree centrality, betweenness centrality, network connectivity measures	The study focuses on general information spread and does not specifically address propaganda detection or classification.	Supports the network analysis component of the project, especially in identifying hubs and bridges and understanding propagation behaviour.

3. Identification of Research Gaps/ Limitations of Existing Models

From the reviewed literature, several important limitations and research gaps can be identified in the existing approaches to propaganda detection and analysis.

Firstly, most existing studies tend to focus on either content-based detection or network-based propagation analysis, but rarely integrate both aspects into a single framework. While text classification models can identify propaganda effectively, they do not provide insight into how such content spreads across social networks. Similarly, network-based studies analyze information diffusion but often do not incorporate content-level understanding.

Secondly, traditional machine learning models such as Logistic Regression rely heavily on surface-level features and lack the ability to capture deep contextual and semantic meaning in text. This limits their effectiveness when dealing with complex language patterns, subtle persuasion techniques, or context-dependent propaganda.

Although transformer-based models such as BERT and RoBERTa significantly improve classification performance by capturing contextual relationships, they are typically used only for text classification tasks. Their integration with network diffusion models is still limited in existing research, resulting in a gap between accurate detection and realistic propagation analysis.

Furthermore, many studies address detection and mitigation as separate problems. Detection-focused research aims to classify content, while mitigation-focused work explores strategies to control spread. However, very few approaches provide a unified system that combines detection, propagation simulation, and intervention analysis within a single platform.

Another notable limitation is the lack of interactive and visual tools for understanding propagation dynamics. Most studies present results in the form of static graphs or numerical outputs, which makes it difficult for users to intuitively understand how propaganda spreads over time and how different factors influence this process.

These gaps highlight the need for a comprehensive and integrated system that combines NLP-based classification, graph-based propagation modeling, and simulation-driven visualization. Such an approach can provide deeper insights into both the nature of propaganda and its behavior within complex social networks.

Chapter 3: System Design and Methodology

1. Proposed Methodology

The proposed system is designed as an integrated analytical framework for studying the propagation and mitigation of propaganda in online social networks. It combines techniques from Natural Language Processing (NLP), Machine Learning, Deep Learning, and Social Network Analysis (SNA) to address two key aspects: accurate detection of propaganda content and simulation of its spread within a network.

The overall methodology is structured into three major components, each responsible for a specific stage of analysis.

(A) Propaganda Detection Module

The first component focuses on identifying whether a given input text contains propaganda. This stage plays a crucial role, as accurate classification forms the basis for further propagation analysis.

The input text undergoes a preprocessing phase, which includes cleaning, normalization, and tokenization to ensure consistency and remove noise. After preprocessing, the text is transformed into numerical representations using two complementary approaches:

- **TF-IDF (Term Frequency–Inverse Document Frequency):**
This method captures the importance of words based on their frequency and relevance within the dataset, making it suitable for traditional machine learning models.
- **Transformer-based embeddings (BERT and RoBERTa):**
These models generate contextual representations of text, allowing the system to understand semantic meaning, word dependencies, and nuanced language patterns.

Three classification models are trained and utilized:

- **Logistic Regression:** A baseline model that performs efficiently on TF-IDF features and provides fast predictions.

- **BERT:** A deep learning model capable of capturing contextual relationships within text.
- **RoBERTa:** An optimized variant of BERT that enhances performance through improved training strategies.

To improve overall reliability, a soft voting ensemble technique is applied. Instead of relying on a single model, the system combines prediction probabilities from all three models to generate the final output. This approach reduces individual model bias and results in more stable and accurate classification performance.

(B) Social Network Construction & Propagation Modeling

Once the text is classified as propaganda, the system proceeds to simulate its spread across a social network.

The network is represented as a graph structure, where:

- Nodes represent users, and
- Edges represent interactions or connections between users.

To mimic real-world social networks, a Barabási–Albert (BA) scale-free network is generated. This model reflects the presence of a few highly connected nodes and many low-degree nodes, which is a common characteristic of social media platforms.

To analyze the structural importance of nodes within the network, the system uses:

- **Degree Centrality:** Identifies highly connected nodes (hubs) that can influence a large number of users.
- **Betweenness Centrality:** Identifies bridge nodes that connect different communities and enable cross-network information flow.

The propagation of propaganda is modeled using the SI (Susceptible–Infected) model, a widely used diffusion model. In this framework:

- **Susceptible (S):** Users who have not yet been exposed to propaganda
- **Infected (I):** Users who have adopted and are spreading propaganda

The transition from susceptible to infected occurs probabilistically, governed by an infection rate parameter (β). This allows controlled simulation of how rapidly propaganda spreads under different conditions.

(C) Mitigation Strategy Module

To study how the spread of propaganda can be controlled, the system incorporates a dedicated mitigation module within the simulation.

This module applies targeted intervention strategies to modify the network and observe their impact on propagation dynamics. The key strategies include:

- **Hub Node Removal:**
Highly connected nodes are deactivated to reduce large-scale and rapid dissemination.
- **Bridge Node Removal:**
Nodes that connect different communities are deactivated to prevent cross-community spread.
- **Infection Probability Control (β Adjustment):**
The propagation rate is reduced for selected nodes to simulate decreased influence or resistance to spreading propaganda.

These strategies allow the system to analyze how network structure and node importance influence propagation behavior, and how targeted interventions can effectively limit the spread.

(D) Design Diagrams

1. Use Case Diagram

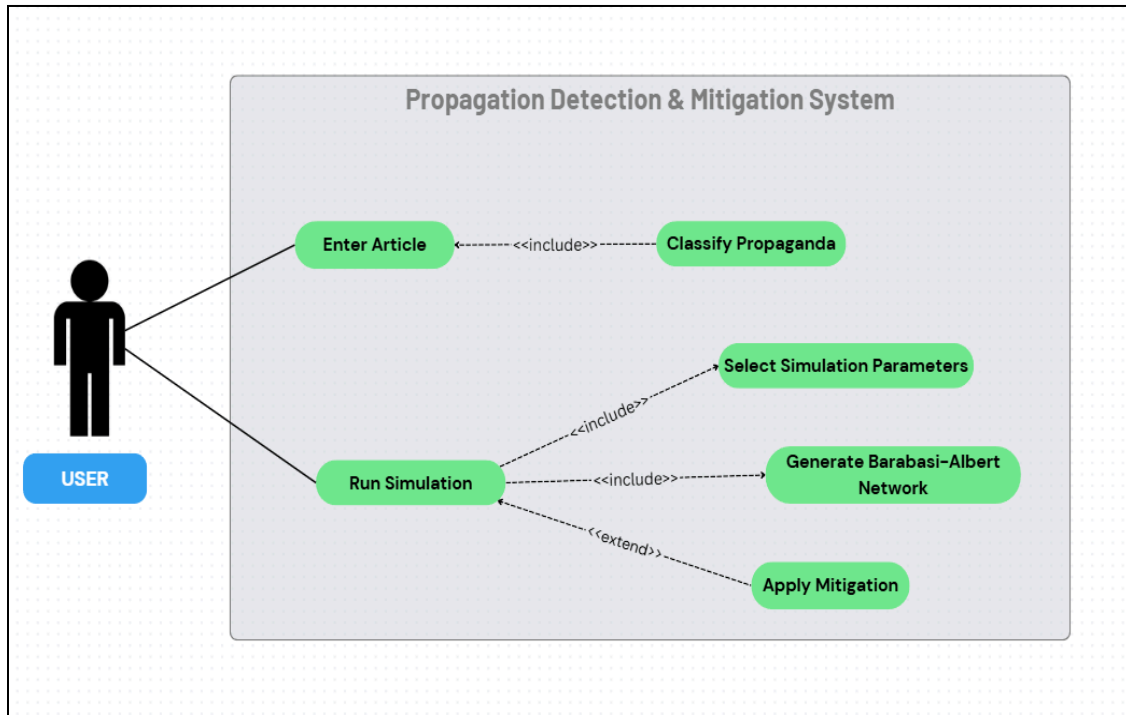


Fig 1: Workflow of user text classification and propaganda spread simulation.

The figure shows how a user inputs text for classification and, if propaganda is detected, how the system proceeds to simulate its spread through various network modes.

2. Flow Chart

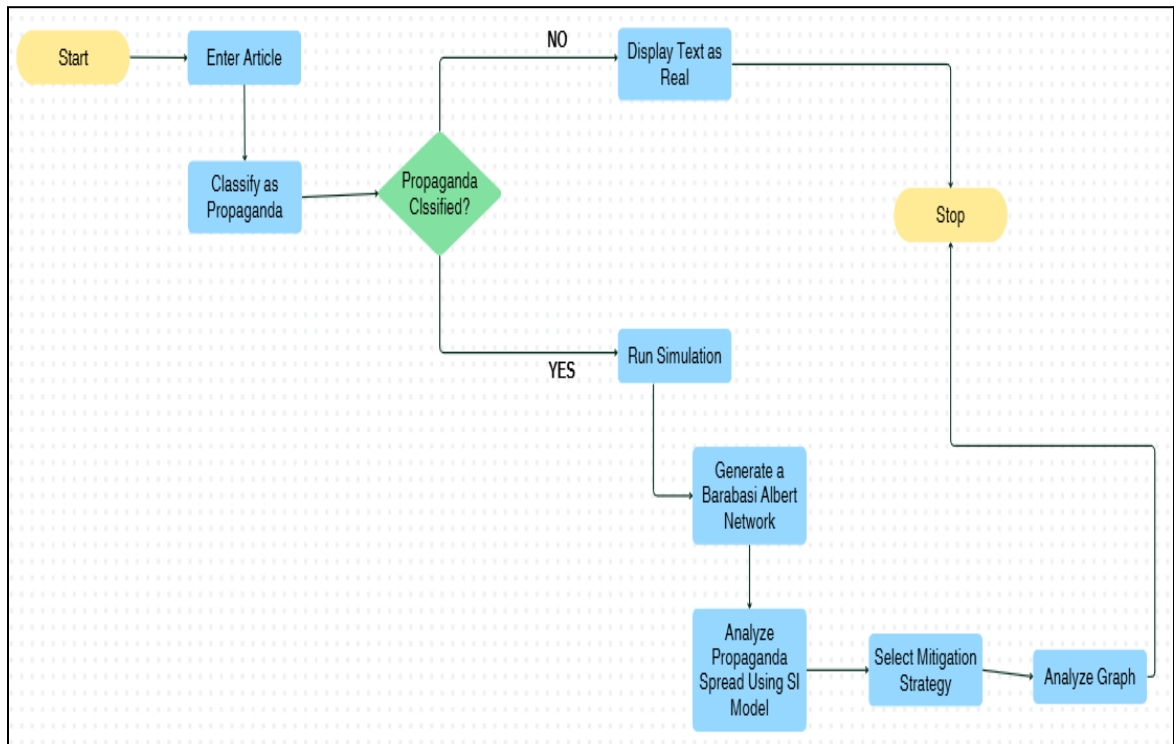


Fig 2: Flowchart of propaganda detection and spread simulation.

The flowchart shows how user text is classified for propaganda and, if detected, how the system simulates and visualizes its spread and mitigation through a selected network model.

3. Class Diagram

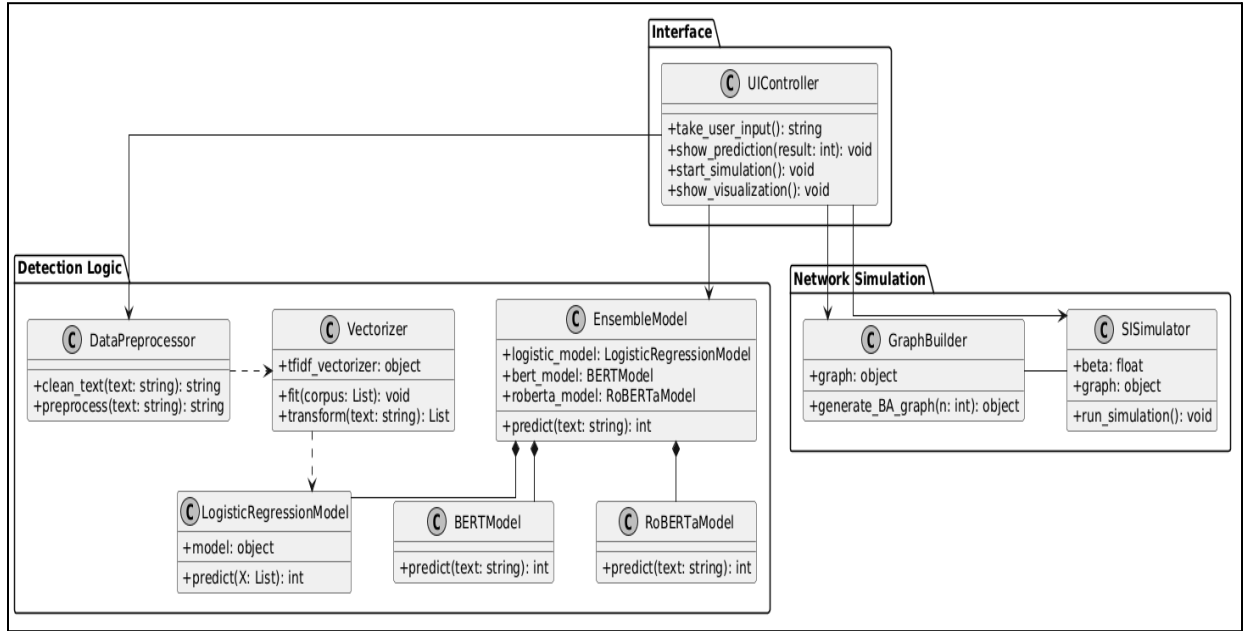


Fig 3: Class diagram of the system's core modules and interactions.

This class diagram presents the core modules of the system and their interactions across preprocessing, feature extraction, machine-learning classification, graph construction, and SI-based propaganda simulation.

2. Tool and Technique Selection

The selection of tools and techniques is based on performance, scalability, interpretability, and suitability for hybrid modeling of text classification and network simulation.

(A) Machine Learning & NLP Techniques

- **Logistic Regression**

Chosen as a baseline model due to its efficiency and strong performance on high-dimensional TF-IDF features.

- **BERT (Bidirectional Encoder Representations from Transformers)**

Used for deep contextual understanding of text, capturing semantic relationships beyond

surface-level features.

- **RoBERTa (Robustly Optimized BERT Pretraining Approach)**

Selected for improved training stability and better contextual representation compared to standard BERT.

- **Soft Voting Ensemble**

Combines predictions from all models to improve robustness and reduce individual model bias.

(B) Feature Extraction Techniques

- **TF-IDF Vectorization**

Captures frequency-based importance of words and works effectively with classical ML models.

- **Transformer Embeddings (BERT/RoBERTa)**

Capture deep contextual and semantic meaning in text.

(C) Network Analysis Techniques

- **NetworkX**

Used for constructing and analyzing social network graphs.

- **Centrality Measures**

- Degree Centrality → identifies hub nodes
- Betweenness Centrality → identifies bridge nodes

These metrics help determine influential nodes in propagation.

(D) Propagation Model

- **SI (Susceptible–Infected) Model**

Selected to simulate propaganda diffusion where users transition from unaware to active spreaders of propaganda. It effectively models continuous propagation without recovery, suitable for studying uncontrolled propaganda spread.

(E) Visualization Tools

- **PyVis**

Used for interactive and dynamic visualization of network propagation.

- **NetworkX Visualization Layer**

Supports structural graph representation and simulation updates.

(F) Programming & Frameworks

- Python: core implementation language
- Scikit-learn: classical ML models
- PyTorch: deep learning framework
- Transformers library: BERT and RoBERTa models
- Pandas / NumPy: data handling and preprocessing

Chapter 4: Implementation and Testing

4.1 Experimental Setup

The experimental setup defines the environment, tools, datasets, and system configuration used for implementing and evaluating the proposed propaganda detection and propagation simulation system.

The system is developed in Python and integrates multiple machine learning, deep learning, and graph-based libraries to achieve a hybrid analytical framework.

4.1.1 System Environment

The implementation is carried out on a standard computing environment with the following specifications:

- Python 3.x as the primary programming language
- Minimum 8GB RAM recommended for transformer-based model execution
- CPU-based execution with optional GPU acceleration for BERT and RoBERTa models

4.1.2 Libraries and Tools Used

The system uses the following libraries:

- **pandas:** data loading and preprocessing
- **scikit-learn:** TF-IDF vectorization and Logistic Regression
- **PyTorch & Transformers:** implementation of BERT and RoBERTa models
- **NetworkX:** graph construction and centrality analysis
- **PyVis:** interactive network visualization
- **NumPy:** numerical computation and array processing

4.1.3 Dataset Description

The dataset used in this project is based on an Indian socio-linguistic context, consisting of textual content originally available in Hindi language. Since the classification models (especially transformer-based models such as BERT and RoBERTa) perform more effectively on English text, the dataset was first translated into English using appropriate translation techniques.

Following translation, the dataset underwent a structured preprocessing pipeline to ensure data quality and consistency. The preprocessing steps included:

- Removal of null and missing values
- Elimination of duplicate entries
- Text normalization and cleaning

After preprocessing, the final dataset was structured and prepared for model training.

The resulting dataset contains a total of 2661 samples, distributed as follows:

- Propaganda samples: 1352
- Non-propaganda samples: 1309

This balanced dataset was then used for training and evaluating all classification models, including Logistic Regression, BERT, RoBERTa, and the proposed soft voting ensemble model.

4.1.4 System Workflow Overview

1. The proposed system follows a structured and modular pipeline that integrates text classification with network-based propagation analysis.
2. Initially, the input text is provided by the user through the interface. This text can be in the form of an article, statement, or any content that needs to be analyzed for propaganda.
3. The input then undergoes a preprocessing stage, where it is cleaned and normalized. This includes converting text to lowercase, removing unnecessary characters, and performing tokenization to break the text into meaningful units. These steps ensure that the data is in a consistent format suitable for further processing.
4. Next, feature extraction is performed using both traditional and advanced techniques.

TF-IDF is used to capture the importance of words in the text, while transformer-based embeddings (from BERT and RoBERTa) are used to capture contextual meaning and semantic relationships.

5. The processed features are then passed through multiple classification models, including Logistic Regression, BERT, and RoBERTa. Each model independently predicts whether the input text is propaganda or not. These predictions are combined using a soft voting ensemble method, which aggregates the confidence scores of all models to produce a more stable and accurate final prediction.
6. If the content is classified as propaganda, the system proceeds to the propagation analysis phase. In this stage, an SI-based network simulation is triggered to model how the identified propaganda may spread across a social network.
7. Finally, the system provides network visualization, where the propagation process is displayed dynamically. Nodes change states to represent the spread of propaganda, and mitigation strategies such as hub or bridge removal can be applied to observe their impact on controlling the spread.
8. This modular workflow ensures that each component operates efficiently while contributing to a comprehensive analysis of both detection and propagation.

4.1.5 Network and Simulation Environment

The system utilizes a Barabási–Albert (BA) scale-free network model to simulate real-world social network structures. This type of network is chosen because it reflects common characteristics of social media platforms, where a few nodes (users) have a very high number of connections, while most nodes have relatively fewer connections. In this representation, nodes correspond to users, and edges represent interactions or connections between them.

To model the spread of propaganda, the system employs the SI model, which is widely used for simulating information diffusion. In this model, each node exists in one of two states: susceptible (not yet exposed to propaganda) or infected (actively spreading propaganda). The transition from susceptible to infected occurs probabilistically, based on a defined infection rate parameter (β). This allows controlled simulation of how quickly or slowly propaganda spreads across the network.

To analyze and control the spread, the system incorporates several mitigation strategies. These include:

- **Hub node removal**, where highly connected nodes are deactivated to reduce rapid spread,
- **Bridge node removal**, where nodes connecting different communities are targeted to limit cross-community propagation, and
- **Infection probability control**, where the value of β is adjusted to simulate reduced influence or spreading capability.

These strategies enable users to observe how different interventions impact the overall propagation behavior, making the simulation both analytical and interactive in nature.

4.2 Test Cases/ Case Study/ Outputs

This section presents the testing strategy, validation process, and observed outputs of the proposed system. The objective is to ensure that each module functions correctly both independently and as part of the integrated system. Multiple levels of testing were performed, including unit testing, integration testing, system testing, and user acceptance testing.

4.2.1 Unit Testing

Unit testing was conducted to evaluate the functionality of individual modules in isolation. Each component was tested to ensure correctness and stability before integration.

- **Text Preprocessing Module:**
Verified operations such as cleaning, normalization, and tokenization to ensure consistent and structured input.
- **Classification Models:**
Evaluated the performance and output format of Logistic Regression, BERT, and RoBERTa models to ensure valid and interpretable predictions.
- **Graph Generation Module:**
Confirmed the correct creation of network structures, including node count, connectivity,

and scale-free properties.

- **SI Simulation Module:**

Validated the correctness of state transitions (Susceptible → Infected) and ensured proper handling of infection probability.

4.2.2 Integration Testing

Integration testing ensured that all modules interact seamlessly and that data flows correctly across the system pipeline.

- Verified the complete flow from preprocessing → classification → prediction output
- Ensured that classification results correctly trigger the simulation module when propaganda is detected
- Confirmed proper linkage between network generation and SI-based propagation
- Validated that mitigation strategies correctly influence the simulation behavior

4.2.3 System Testing

System testing was performed to evaluate the performance of the entire system under realistic conditions.

- Classification of input text as propaganda or non-propaganda
- Automatic triggering of SI-based propagation simulation
- User-driven selection of hub-based or bridge-based propagation
- Visualization of propagation across the network
- Observation of the impact of mitigation strategies on spread behavior

This stage ensured that the system operated correctly from input to final output.

4.2.4 User Acceptance Testing (UAT)

User-level testing was conducted to evaluate usability, clarity, and overall interaction with the system.

- Ease of entering and submitting input text
- Clarity and interpretability of classification results
- Understanding of propagation visualization
- Ease of using simulation controls such as hub/bridge selection

Feedback from this stage helped validate the system from an end-user perspective.

4.2.5 Sample Test Cases

A set of representative test cases was used to validate key functionalities:

- **Test Case 1: Text Classification**
Input: Propaganda and non-propaganda samples
Expected Output: Correct classification label
Result: Ensemble model produced consistent and accurate predictions
- **Test Case 2: SI Propagation Simulation**
Input: Selected initial infected node
Expected Output: Gradual propagation across the network
Result: Nodes transitioned correctly from Susceptible to Infected
- **Test Case 3: Mitigation Effect**
Input: Application of hub/bridge deactivation strategies
Expected Output: Reduction in propagation spread
Result: Significant decrease in infection spread observed
- **Test Case 4: Visualization Output**
Input: Simulation execution
Expected Output: Dynamic network graph representation
Result: Real-time node color changes accurately reflected state transitions
- **Test Case 5: Error Handling**
Input: Empty/invalid text or incorrect node selection
Expected Output: Proper error message
Result: System handled exceptions gracefully without failure

4.2.6 Observed Outputs (System Behavior)

The following observations were recorded during testing:

- Logistic Regression demonstrated efficient performance on TF–IDF features
- BERT and RoBERTa improved contextual understanding and classification accuracy
- The ensemble model provided more stable and balanced predictions compared to individual models
- The SI model effectively simulated propagation dynamics across the network
- Mitigation strategies significantly reduced the spread under controlled conditions
- PyVis-based visualization clearly illustrated the flow of propagation

4.2.7 Key Observations from Testing

Based on the overall evaluation, the following key insights were derived:

- The ensemble approach enhances robustness compared to individual models
- Network structure plays a critical role in determining propagation speed
- Hub nodes act as major spreaders, accelerating propagation
- Bridge nodes enable cross-community spread, increasing reach
- Mitigation strategies are effective in controlling and limiting propagation intensity

Chapter 5: Results and Discussion

5.1 Result Analysis

The experimental evaluation of the proposed system is divided into two major components:

1. Propaganda Classification Performance (ML + DL Models)
2. Propagation Behavior Analysis using Social Network Simulation (SI Model)

5.1.1 Classification Model Evaluation

The performance of the classification models (Logistic Regression, BERT, RoBERTa, and Soft Voting Ensemble) is evaluated using standard classification metrics:

Accuracy : The ratio of correctly predicted observations to the total observations. It measures how often the model is correct.

$$\text{Accuracy} = \frac{\text{True Positives} + \text{True Negatives}}{\text{True Positives} + \text{True Negatives} + \text{False Positives} + \text{False Negatives}}$$

Fig 4: Formula for Accuracy

Precision: It measures how many of the "positive" predictions made by the model were actually correct.

$$\text{Precision} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}}$$

Fig 5: Formula for Precision

Recall: It measures how many of the actual positive cases the model was able to identify correctly.

$$\text{Recall} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}}$$

Fig 6: Formula for Recall

F1 Score: It is the harmonic mean of Precision and Recall. It is used to find a balance between the two, especially when class distribution is uneven.

$$\text{F1 Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

Fig 7: Formula for F1-score

Observations:

- Logistic Regression

Test Classification Report				
	precision	recall	f1-score	support
0.0	0.87	0.85	0.86	271
1.0	0.84	0.87	0.85	262
accuracy			0.86	533
macro avg	0.86	0.86	0.86	533
weighted avg	0.86	0.86	0.86	533

Fig 8: Accuracy, Precision, Recall and F1-score for Logistic Regression Model

- BERT

	precision	recall	f1-score	support
Non-Propaganda	0.86	0.82	0.84	262
Propaganda	0.83	0.87	0.85	271
accuracy			0.85	533
macro avg	0.85	0.85	0.85	533
weighted avg	0.85	0.85	0.85	533

Fig 9: Accuracy, Precision, Recall and F1-score for BERT

- RoBERTa

===== TESTING SET REPORT =====				
	precision	recall	f1-score	support
Non-Propaganda	0.84	0.82	0.83	131
Propaganda	0.83	0.85	0.84	136
accuracy			0.84	267
macro avg	0.84	0.84	0.84	267
weighted avg	0.84	0.84	0.84	267

Fig 10: Accuracy, Precision, Recall and F1-score for RoBERTa

- Ensemble Model

=====				
FINAL ENSEMBLE REPORT				
=====				
	precision	recall	f1-score	support
Non-Propaganda	0.85	0.81	0.83	131
Propaganda	0.82	0.86	0.84	136
accuracy			0.84	267
macro avg	0.84	0.83	0.83	267
weighted avg	0.84	0.84	0.84	267

Fig 11: Accuracy, Precision, Recall and F1-score for the Soft Voting Ensemble

5.1.2 SI Model for Propagation Analysis

The propagation of propaganda is modeled using the **Susceptible–Infected (SI) model**, where nodes transition from susceptible to infected states over time.

Key Observations:

- Propagation is faster when initiated from hub nodes due to high connectivity.
- Bridge nodes enable cross-community spread, increasing reach over time.
- Without mitigation, the infection continues to grow monotonically (no recovery state).

5.2 Performance Evaluation / Comparison with Existing Methods

The proposed system is compared conceptually with traditional approaches used in literature.

5.2.1 Classification Performance Comparison

Table 3: Performance Evaluation of Classification Models

Method	Feature Type	Context Understanding	Performance
Logistic Regression	TF-IDF	Low	Moderate
BERT	Contextual embeddings	High	High
RoBERTa	Optimized contextual embeddings	Very High	Higher
Proposed Ensemble Model	TF-IDF + Transformers	Very High	Best (Balanced)

Key Insight

The ensemble approach outperforms individual models by combining:

- lexical feature strength (Logistic Regression)
- contextual understanding (BERT, RoBERTa)

This results in higher robustness and reduced misclassification rate.

5.2.2 Propagation Model Evaluation

Table 4: Comparison of various propagation models

Approach	Network Awareness	Realism	Control Ability
Rule-based spread models	Low	Low	Low
Basic diffusion models	Medium	Medium	Limited
SI Model	High (graph-based)	High	Medium
SI + Mitigation (Proposed System)	Very High	Very High	High

Key Insight

- Hub-based propagation results in rapid exponential-like spread
- Bridge-based propagation results in delayed but wide dissemination
- Mitigation strategies significantly reduce propagation reach

5.2.3 System Performance Observations

- The system performs efficiently on medium-sized networks
- PyVis visualization provides clear real-time propagation tracking
- Larger graphs lead to increased computational load during rendering
- Ensemble classification remains stable across diverse text inputs

5.2.4 Key Findings from Results

- Ensemble model improves classification stability significantly

- Transformer models enhance contextual detection of propaganda
- Network structure plays a crucial role in propagation dynamics
- Hubs act as super-spreaders in social networks
- Bridges enable cross-community propaganda flow
- Mitigation strategies effectively control spread intensity

5.2.5 Visualization-Based Insights

- Early-stage propagation shows localized infection around source nodes
- Mid-stage propagation expands rapidly through hubs
- Final-stage propagation stabilizes after mitigation application

Chapter 6: Conclusion and Future Scope

6.1 Summary of Work and Achievements

The project titled “Propagation and Mitigation of Propaganda in Online Social Networks” presents a comprehensive system that integrates Natural Language Processing (NLP), Machine Learning, Deep Learning, and Social Network Analysis (SNA) to analyze both the detection and propagation of propaganda in online environments.

The primary objective of the system is to develop a unified framework that not only classifies textual content as propaganda or non-propaganda but also simulates its spread across a social network and evaluates the effectiveness of different mitigation strategies. This dual-level approach enables both content-level understanding and network-level analysis of propaganda.

The key achievements of the project include the development of a hybrid classification system using Logistic Regression, BERT, and RoBERTa models. These models are combined using a soft voting ensemble approach, which improves classification stability and ensures a better balance between precision and recall. The integration of transformer-based models with classical machine learning techniques enhances the system’s ability to capture both surface-level and contextual features of text.

On the network side, the project implements a social network simulation using the Barabási–Albert model, which effectively represents real-world scale-free network structures. The SI (Susceptible–Infected) model is used to simulate how propaganda spreads across the network over time, allowing observation of propagation patterns under different conditions.

The system also includes interactive visualization using NetworkX and PyVis, enabling users to dynamically observe how nodes transition from susceptible to infected states. This visual representation improves understanding of how network structure influences the spread of information.

Furthermore, the project incorporates mitigation strategies, such as hub removal, bridge removal, and control of infection probability, which allow users to analyze how targeted interventions can

reduce or slow down propagation. This makes the system not only analytical but also exploratory in nature.

Overall, the project results in the creation of a simulation-based analytical tool similar to Gephi, designed for educational and research purposes. Although the project does not include formal publications or external awards at this stage, it serves as a strong academic implementation that combines multiple domains of artificial intelligence and network science in a meaningful and practical way.

6.2 Impact on Society, Environmental Sustainability, Ethical Issues & Industry Relevance

(A) Social Impact

The system directly addresses the growing issue of propaganda in online social networks. Its ability to detect and simulate propaganda spread can help in:

- Understanding how propaganda influences public opinion
- Identifying critical nodes responsible for rapid spread
- Supporting awareness of digital propaganda patterns
- Assisting in designing better content moderation strategies

This makes the system valuable for researchers, educators, and policy analysts working in the domain of digital safety.

(B) Environmental Sustainability

Although the system is computational in nature, it contributes indirectly to sustainability by:

- Reducing dependency on manual content moderation systems
- Enabling automated detection and simulation of propagation patterns
- Optimizing decision-making processes in large-scale social media analysis

Efficient model design and ensemble optimization also help reduce redundant computations during training and inference.

(C) Ethical Considerations

The project handles sensitive content classification, which introduces several ethical considerations:

- The system is designed strictly for research and educational purposes, not for surveillance or censorship.
- Bias in training data may influence classification outcomes, requiring careful dataset selection.
- Misclassification risks are acknowledged, especially for ambiguous or context-dependent content.
- The system avoids real-time user monitoring and operates only on provided input data.

Ethical use guidelines ensure that the system promotes responsible AI development and transparency.

(D) Industry Relevance

The techniques used in this project are highly relevant to modern industry applications such as:

- Social media content moderation systems
- Propaganda detection platforms
- Network analytics tools for user influence tracking
- Digital risk analysis systems for online platforms

The combination of transformer models, graph-based simulation and ensemble learning reflects real-world AI system design patterns used in industry-scale applications.

6.3 Limitations of the Work

Despite achieving effective and meaningful results, the proposed system has certain limitations that should be considered:

1. Dependence on Dataset Quality:

The performance of the classification models is highly dependent on the quality, diversity, and representativeness of the training dataset. If the dataset does not adequately capture different styles or types of propaganda, the model's ability to generalize to unseen data may be affected.

2. Limited Handling of Informal and Complex Text:

The system may struggle with text containing slang, sarcasm, emojis, or highly contextual and ambiguous expressions. Such variations are difficult to interpret accurately, even for advanced models, which can lead to occasional misclassification.

3. Simplified Propagation Model:

The SI (Susceptible–Infected) model used for simulation assumes that once a node becomes infected, it remains infected permanently. This does not account for recovery or loss of interest, making the long-term propagation dynamics less realistic compared to real-world scenarios.

4. Synthetic Network Structure:

The use of Barabási–Albert graphs provides a reasonable approximation of scale-free networks; however, it does not fully capture the complexity, dynamics, and evolving nature of real social media platforms.

5. Scalability Constraints:

While the system performs well on small to medium-sized networks, larger networks can lead to increased computational load and slower visualization, affecting real-time interaction and user experience.

6. Static Analysis Environment:

The system operates on pre-loaded datasets and does not incorporate real-time streaming data from live platforms. As a result, it cannot reflect continuously changing trends or emerging propaganda in real time.

7. Limited External Validation:

The system has been evaluated in a controlled academic environment. Its performance in real-world deployment scenarios may vary due to factors such as noisy data, dynamic user behavior, and platform-specific characteristics.

6.4 Future Scope

Several enhancements can be implemented to further extend the capabilities and real-world applicability of the proposed system:

(A) Advanced Transformer Models:

Future improvements can include the integration of lightweight and optimized transformer architectures such as DistilBERT, ALBERT, or DeBERTa. These models can reduce computational cost while maintaining high classification accuracy, making the system more efficient and suitable for real-time applications.

(B) Real-World Social Media Integration:

The system can be extended to incorporate real-time data from platforms such as Twitter, Reddit, or other social networks. This would enable simulation of live propaganda propagation scenarios and provide more realistic insights into how propaganda spreads in dynamic environments.

(C) Advanced Diffusion Models:

Currently, the system uses the SI model, which is relatively simple. Future work can include more advanced diffusion models such as SIR (Susceptible–Infected–Recovered), SIS, and opinion dynamics models. These models can better represent real-world behavior by accounting for recovery, repeated exposure, and varying user responses.

(D) Multi-Source and Coordinated Propagation:

The system can be enhanced to simulate multiple simultaneous sources of propagation, coordinated propaganda campaigns, and bot-driven spread patterns. This would improve the realism of the simulation and allow analysis of more complex and large-scale scenarios.

(E) Scalability and Performance Optimization:

To support larger networks, performance can be improved using techniques such as parallel processing, GPU-accelerated graph computations, and optimized visualization pipelines. These enhancements would enable smoother execution and faster analysis of large-scale simulations.

(F) Real-Time Monitoring System:

The system can be developed into a real-time monitoring tool capable of tracking live

propagation trends, detecting emerging propaganda campaigns, and providing dynamic alerts. This would significantly increase its practical relevance in real-world applications.

(G) Enhanced Explainability and Trust:

Future work can incorporate more advanced Explainable AI techniques to provide deeper insights into model decisions. This would improve transparency in ensemble predictions and help build greater user trust in automated classification systems.

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Appendices

Appendix A: Dataset Description

The dataset used in this project consists of textual news and social media content labeled as propaganda or non-propaganda.

Dataset Features:

- Text content (news articles)
- Label (0 = Non-Propaganda, 1 = Propaganda)

Preprocessing Steps:

- Translation of the dataset from Hindi to English
- Removal of special characters and punctuation
- Lowercasing of text
- Tokenization of sentences
- Stopword removal (where applicable)
- Conversion into TF-IDF and transformer embeddings

Final Dataset Usage:

The cleaned dataset is used for training:

- Logistic Regression model (TF-IDF features)
- BERT model (contextual embeddings)
- RoBERTa model (enhanced contextual embeddings)

Appendix B: Key Code Snippets

1. TF-IDF Feature Extraction

```
from sklearn.feature_extraction.text import TfidfVectorizer

vectorizer = TfidfVectorizer(
    max_features=5000,
    ngram_range=(1,2),
    stop_words='english'
)

X_train_vec = vectorizer.fit_transform(X_train)
X_test_vec = vectorizer.transform(X_test)
```

Fig 12: Initializing TF-IDF Vectorizer for Feature Extraction

2. Logistic Regression Model

```
from sklearn.linear_model import LogisticRegression

best_model = LogisticRegression(max_iter=1000)
best_model.fit(X_train_vec, y_train)
```

Fig 13: Training Logistic Regression Model

3. BERT Pipeline

```
tokenizer = BertTokenizer.from_pretrained("bert-base-uncased")

def tokenize(texts):
    return tokenizer(
        texts,
        padding=True,
        truncation=True,
        max_length=512
    )

train_enc = tokenize(train_df["text"].tolist())
val_enc   = tokenize(val_df["text"].tolist())
```

Fig 14: BERT Tokenizer

```
bert_model = BertForSequenceClassification.from_pretrained(
    "bert-base-uncased",
    num_labels=2
)
```

Fig 15: Initializing BERT for training

4. RoBERTa Pipeline

```
# tokenizaion
model_name = "roberta-base"
tokenizer = AutoTokenizer.from_pretrained(model_name)

def tokenize_fn(batch):
    return tokenizer(batch["text"], padding="max_length", truncation=True, max_length=512)

tokenized_datasets = dataset.map(tokenize_fn, batched=True)

# model + hyperparams
model = AutoModelForSequenceClassification.from_pretrained(model_name, num_labels=2)

training_args = TrainingArguments(
    output_dir="./roberta_propaganda_model",
    eval_strategy="epoch",
    save_strategy="epoch",
    learning_rate=5e-6,
    per_device_train_batch_size=8,
    per_device_eval_batch_size=8,
    num_train_epochs=5,
    weight_decay=0.1,
    load_best_model_at_end=True,
    metric_for_best_model="eval_loss",
    greater_is_better=False,
    logging_steps=20,
    report_to="none"
)
```

Fig 16: RoBERTa Tokenizer and Model Initialisation

5. Ensemble Model

```
def run_ensemble():
    lr_probs = logistic_model.predict_proba(X_test_lr)

    def get_transformer_probs(model, tokenizer, texts):
        model.eval()
        inputs = tokenizer(texts, padding=True, truncation=True, max_length=512, return_tensors="pt")
        with torch.no_grad():
            logits = model(**inputs).logits
        return softmax(logits.numpy(), axis=1)

    bert_probs = get_transformer_probs(bert_model, bert_tokenizer, test_texts)
    roberta_probs = get_transformer_probs(roberta_model, roberta_tokenizer, test_texts)

    ensemble_probs = (lr_probs + bert_probs + roberta_probs) / 3
    final_preds = np.argmax(ensemble_probs, axis=1)

    return final_preds
```

Fig 17: Initializing the Soft Voting Ensemble

6. SI Propagation Model

```
def si_step(G, status, beta, propagation):
    degs = dict(G.degree())
    vals = list(degs.values())
    mean_deg = np.mean(vals)
    median_deg = np.median(vals)
    new_status = status.copy()
    for node in G.nodes():
        if status[node] != 1:
            continue
        for nei in G.neighbors(node):
            if status[nei] != 0:
                # skip already infected (1) and immune (2)
                continue
            spread_ok = True
            if propagation == "Hub":
                spread_ok = degs[nei] > mean_deg
            elif propagation == "Bridge":
                spread_ok = degs[nei] < median_deg
            if spread_ok and random.random() < beta:
                new_status[nei] = 1
    return new_status
```

Fig 18: Applying SI Propagation Model

7. Network Generation

```
def build_graph(n, m=2):  
    return nx.barabasi_albert_graph(n, m, seed=42)
```

Fig 19: Barabasi-Albert Network Generation

Appendix C: Output Screenshots

1. Processed Dataset

	text	label
4342	a special bench of chief justice nv ramana jus...	1.0
4344	congress up congress will launch womens manife...	1.0
4345	ecofriendly fireworks in the golden temple see...	0.0
4347	kisan morcha agreed but kisan union did not po...	1.0
4348	railway news if you are a senior citizen and h...	0.0

Fig 20: Overview of Dataset After Preprocessing

The figure above shows the dataset overview after applying proper preprocessing steps like handling Null values/ duplicates.

2. Ensemble Classifications

Test Case Text	Actual Label	Model Prediction
new delhi december after many years of demand and discussion the post of chief of defense staff was created in...	NON-PROPAGANDA	NON-PROPAGANDA
omicron in india the omicron variant of corona virus is spreading rapidly in the country omicron has also made...	NON-PROPAGANDA	NON-PROPAGANDA
the girl had asked for help after gangrape but the ngo hid the informationxd by deepak kumar pant published de...	NON-PROPAGANDA	NON-PROPAGANDA
national desk chief of defense staff general bipin rawat his wife madhulika rawat and others have died in a he...	NON-PROPAGANDA	NON-PROPAGANDA
uproar over mamita mehar murder case continues for fourth day odisha assembly proceedings adjournedxd ever sin...	PROPAGANDA	PROPAGANDA
chandigarh former punjab chief minister captain amarinder singh said on monday that his partys aim is to win t...	PROPAGANDA	PROPAGANDA
he further said that it is too early to say how contagious it can be the doctor said we dont know how serious ...	NON-PROPAGANDA	NON-PROPAGANDA
pakistan is not desisting from its nefarious activities he keeps planning to harm india in some way or the oth...	NON-PROPAGANDA	NON-PROPAGANDA
indian tennis star leander paes joined tmc on friday paes who has gained international recognition as a tennis...	PROPAGANDA	NON-PROPAGANDA
patna the last day of the winter session of bihar legislative assembly ended with controversy during the concl...	PROPAGANDA	PROPAGANDA
the nia on saturday filed a charge sheet against three people in connection with the bomb blasts outside the r...	PROPAGANDA	PROPAGANDA

Fig 21: Performance of ensemble model on sample test cases

The figure above shows the classifications made by the ensemble model on sample test cases within and outside the dataset.

3. XAI Word Importance Visualization

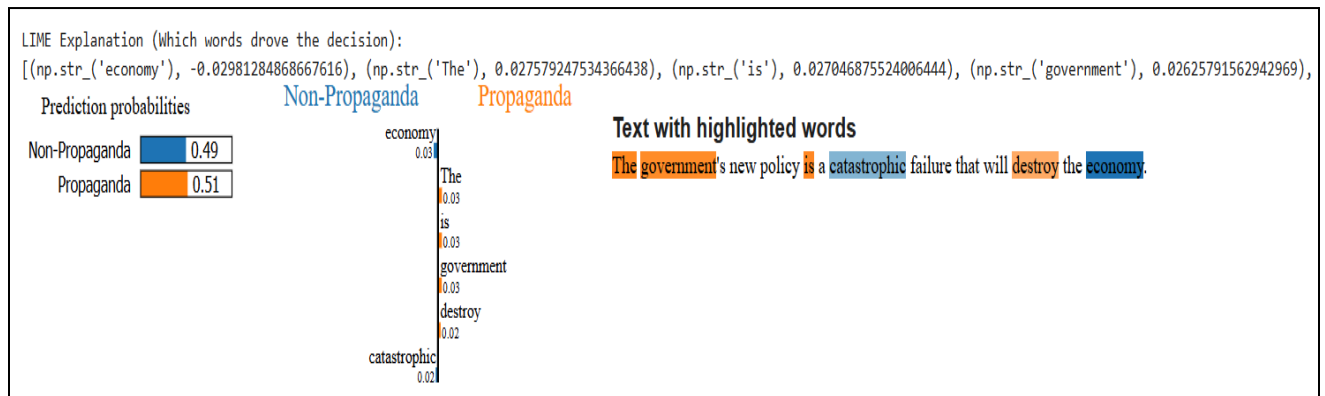


Fig 22: XAI Interpretation for classified text

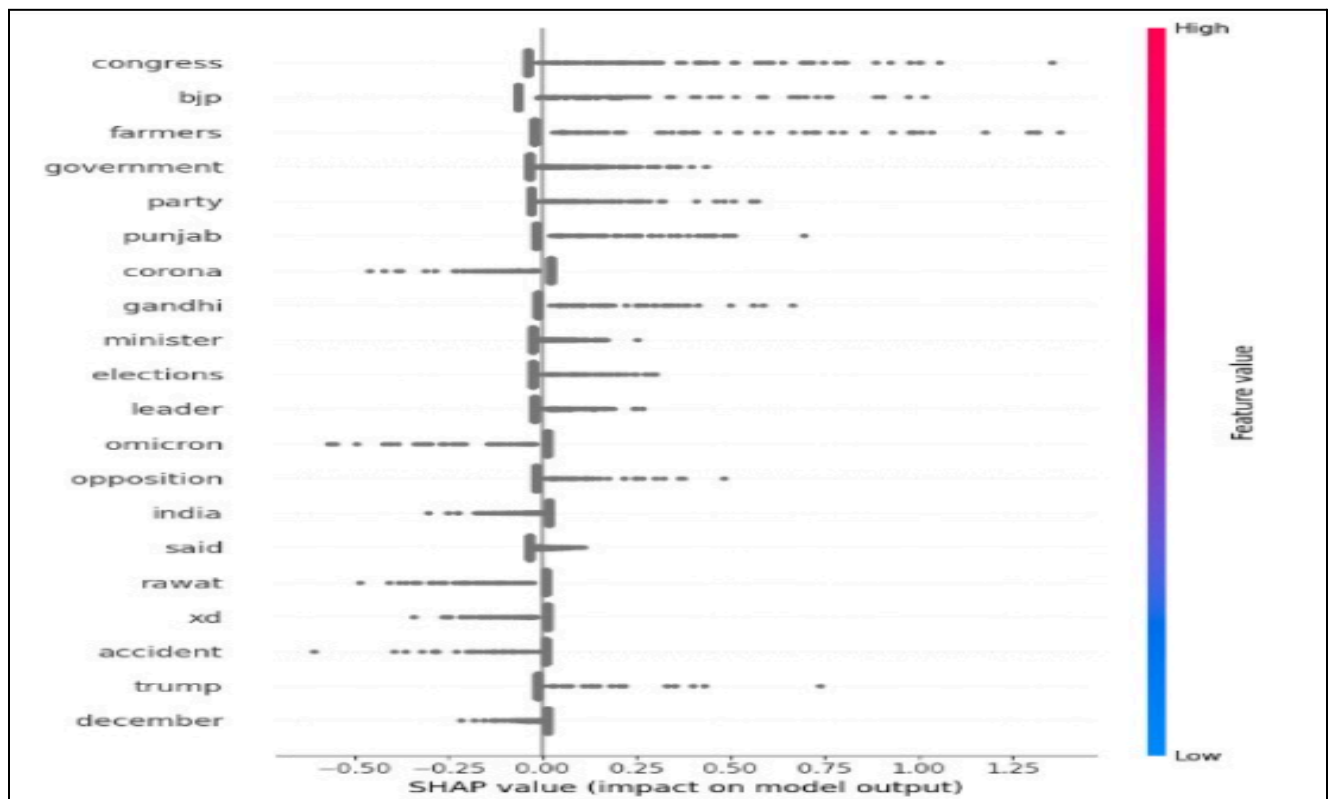


Fig 23: Relative Influence of words in model predictions

Figures 22, 23 above show the relative importance of words in influencing the overall model predictions ensuring transparency in model predictions.

4. User Interface

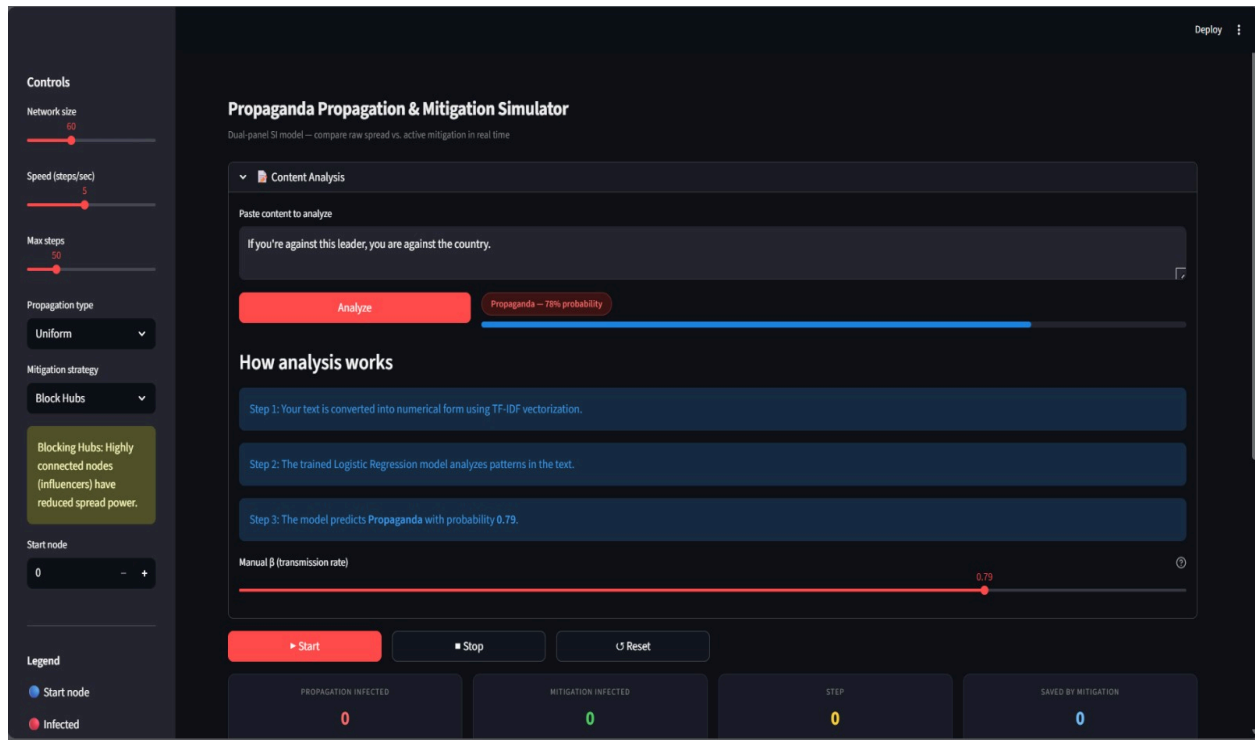


Fig 24: User Interface for Simulation

The figure above illustrates the user interface of the proposed system, enabling users to input textual data for classification, execute and analyze propagation simulations, adjust SI model parameters, and observe the evolving network state at each step for improved interpretability.

5. SI Simulation & Mitigation

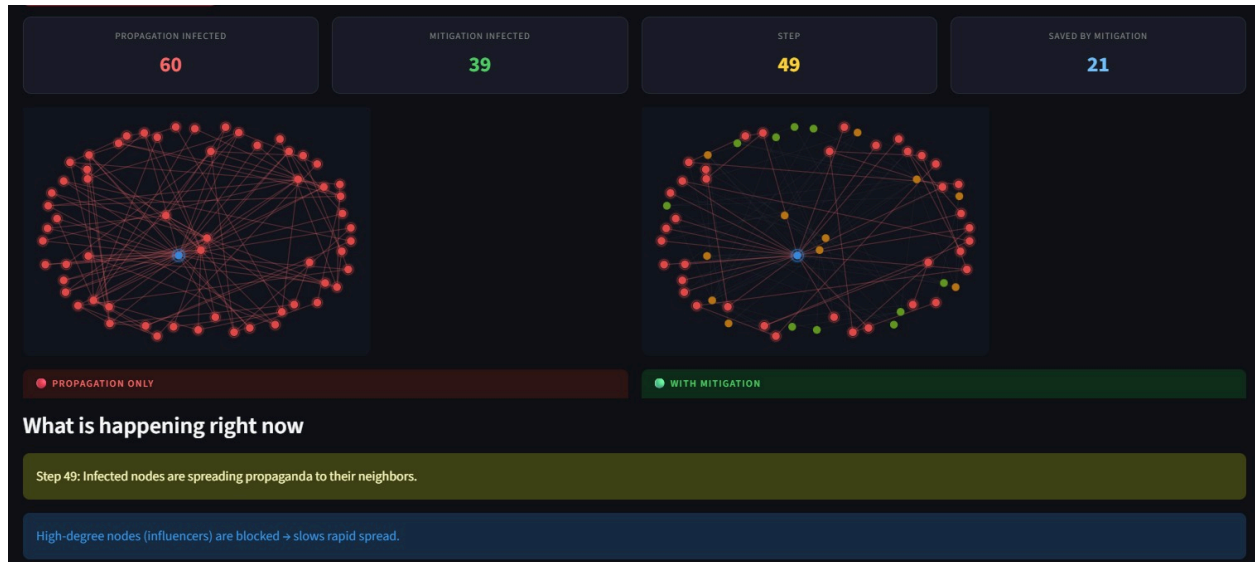


Fig 25: Propagation & Mitigation

The figure above illustrates the network state during SI-based propagation alongside the network after the application of mitigation strategies, highlighting the reduction in spread and structural impact on the network.

Annexure

Annexure A: Outcome of the Project

- The project titled “*Propagation and Mitigation of Propaganda in Social Networks*” successfully delivers a complete system that integrates both propaganda detection and propagation analysis.
- A hybrid classification approach is implemented using Logistic Regression, BERT, and RoBERTa models combined through a soft voting ensemble, which improves overall prediction stability and accuracy. The system is capable of handling different types of textual inputs and provides reliable classification results.
- In addition to detection, the project includes a network-based simulation module using the SI (Susceptible–Infected) model to study how propaganda spreads across a social network. A Barabási–Albert graph is used to represent realistic network structures, allowing meaningful analysis of propagation behavior.
- The system also provides interactive visualization using NetworkX and PyVis, enabling users to observe how information spreads over time. Mitigation strategies such as hub removal and bridge removal are implemented to analyze how the spread of propaganda can be controlled.
- Overall, the project demonstrates a practical integration of NLP, machine learning, and social network analysis, and serves as a useful academic tool for studying propaganda dynamics.

Annexure B: Sustainability Statement

The proposed system contributes to sustainability primarily from a digital and societal perspective, rather than direct environmental impact.

From a social standpoint, the system addresses the growing issue of propaganda in online environments. By enabling the detection and analysis of propaganda content, it supports efforts to promote responsible information consumption and digital awareness. This can indirectly contribute to a more informed society, reducing the harmful effects of misleading or manipulative content.

In terms of operational efficiency, the system reduces reliance on manual content analysis, which is often time-consuming and resource-intensive. Automated classification and simulation allow large volumes of data to be processed efficiently, thereby optimizing computational and human resource utilization.

The use of simulation-based analysis further enhances sustainability by allowing experimentation in a controlled virtual environment. Instead of applying interventions in real-world systems, which may have unintended consequences, the system enables safe testing of mitigation strategies such as node removal or influence control. This reduces risk and promotes responsible decision-making.

Overall, the project contributes to sustainable digital ecosystems by promoting awareness, efficiency, and responsible use of artificial intelligence in tackling propaganda.

Annexure C: Ethical Statement

The development and deployment of the proposed system involve several ethical considerations, particularly due to its focus on content classification and information analysis.

Firstly, the system is designed strictly for academic and research purposes. It does not aim to monitor, track, or control real users or platforms. All inputs are provided manually, and the system does not access or process real-time personal data from social media or other sources.

Secondly, the project acknowledges the presence of dataset bias and limitations. Since the models are trained on specific datasets, their predictions may be influenced by the nature and distribution of the training data. This can lead to misclassification, especially in cases involving sarcasm, satire, or culturally nuanced content. The system does not claim absolute accuracy and should be used with an understanding of these limitations.

Another important ethical consideration is the avoidance of surveillance or misuse. The system does not include any functionality for tracking user behavior, profiling individuals, or performing automated censorship. Its purpose is limited to analysis and simulation in a controlled environment.

Additionally, the integration of Explainable AI (XAI) techniques promotes transparency, allowing users to understand the reasoning behind classification decisions. This helps build trust and ensures that the system does not operate as a “black box.”

Overall, the project adheres to principles of responsible AI usage, ensuring that it is used for learning, research, and awareness rather than control or manipulation.

Annexure D: Team Contributions

Ritika Singh:

- Data collection and translation
- PyVis visualization and UI integration

Shireen Kachroo:

- Data Collection & Preprocessing
- Training Logistic Regression and RoBERTa
- Training the final Soft Voting Ensemble of Logistic Regression, BERT and RoBERTa

Paakhi Somani:

- Training BERT
- SI propagation simulation and UI integration